Fuzzy-Calibrated Conformal Prediction for Adaptive Uncertainty-Aware AI

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Abstract

Reliable uncertainty estimates are essential when machine-learning predictions support high-consequence decisions. This paper presents a transparent fuzzy-calibrated conformal prediction baseline that combines bootstrap logistic ensembles, disagreement-based uncertainty, and adaptive interval scaling. The fuzzy controller maps ensemble disagreement to a bounded uncertainty regime, while conformal calibration preserves finite-sample coverage under exchangeability assumptions. In a seeded experiment using the public scikit-learn breast-cancer dataset as a methodological proxy, the baseline classifier achieved 97.37% accuracy, a Brier score of 0.0210, and expected calibration error of 0.0226. The fuzzy conformal intervals achieved 93.86% empirical coverage with mean width 0.1759 at nominal error level $\alpha = 0.1$. The same construction is extended to selective action selection for autonomous agents, where the agent executes an action only when the conformal prediction set is a singleton and abstains otherwise; on public proxy datasets the agent attains marginal coverage close to the 90% nominal level at low selective risk. These results are not clinical validation; they demonstrate the executable protocol and motivate evaluation on temporal, physiological, and distribution-shift benchmarks.

Keywords

uncertainty quantification, conformal prediction, interval type-2 fuzzy logic, calibration, trustworthy AI

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1 Introduction

Neural and statistical predictors often produce confident outputs outside their training distribution. In high-consequence settings, a useful system must distinguish uncertainty caused by noisy observations from uncertainty caused by insufficient model knowledge. This paper develops a compact, auditable baseline rather than claiming clinical deployment.

2 Related Work

Deep ensembles estimate predictive variability through independently fitted models, while conformal prediction supplies distribution-free finite-sample guarantees under exchangeability. Fuzzy systems provide interpretable rules for translating continuous uncertainty signals into operating regimes. The contribution here is their explicit composition: ensemble disagreement is converted into a bounded fuzzy scale that controls conformal interval width.

3 Methods

Let $f_k(x)$ denote the probability from ensemble member k. The predictive mean and disagreement are:

$$\bar{f}(x) = K^{-1} \sum_{k=1}^{K} f_k(x), \qquad u(x) = \mathrm{std}\{f_k(x)\}_{k=1}^{K}. \quad (1)$$

Calibration scores use the absolute residual $s_i = |y_i - \bar{f}(x_i)|$. For error level α, the conformal radius is the empirical quantile $q_{1-\alpha}$ of calibration scores. The fuzzy controller maps $u(x)$ to $g(x) \in [0, 1]$ and returns:

$$[\ell(x), r(x)] = [\bar{f}(x) - q(1+0.5g(x)), \bar{f}(x) + q(1+0.5g(x))]. \quad (2)$$

The current implementation uses bootstrap logistic-regression members and a transparent quantile-based membership function. It is not an Interval Type-2 Fuzzy Neural Network; that is a planned extension.

4 Experimental Setup

The public scikit-learn breast-cancer dataset is used as a methodological proxy and not as clinical validation. The data are split into 341 training, 114 calibration, and 114 test samples using stratification and seed 20260915. Seven bootstrap ensemble members are trained. We report accuracy, Brier score, expected calibration error, empirical conformal coverage, and mean interval width. No patient-identifying data or clinical decision is involved.

5 Results

Table 1: Seeded proxy benchmark results.

Metric	Observed value
Baseline accuracy	0.9737
Baseline Brier score	0.0210
Baseline ECE	0.0226
Conformal quantile	0.1225
Fuzzy coverage	0.9386
Mean interval width	0.1759

The empirical coverage exceeds the nominal 90% target in this single seeded proxy run. The result does not establish statistical significance or generalization. It confirms that the repository produces a complete uncertainty artifact with separable calibration and test stages.

6 Selective Action Selection for Autonomous Agents

The base construction provides binary coverage. We extend it to safe action selection, where an agent chooses one action from a finite set $\mathcal{A}$ or abstains. For a test instance x the prediction set is

$$\mathcal{C}(x) = \{a \in \mathcal{A} : 1 - p_a(x) \leq \tau(x)\}, \tag{3}$$

where $p_a(x)$ is the ensemble-mean probability of action a. The baseline sets $\tau(x) = q_{1-\alpha}$; the fuzzy variant sets $\tau(x) = q_{1-\alpha}(1 + \beta g(x))$ with $\beta = 0.5$ and $g(x)$ the bounded disagreement map from Eq. (2). The agent executes the action when $|\mathcal{C}(x)| = 1$ and abstains otherwise. We report marginal coverage, mean set size, abstention rate, selective risk (error among executed actions), and the area under the risk–coverage curve (AURC) obtained by ranking instances by top-action confidence.

Fuzzy scaling raises coverage and enlarges prediction sets, trading set size for robustness; on *digits* it also lowered the abstention rate because baseline empty sets were refilled to singletons. The *wine* proxy is near-separable (always-act risk 0.000) and therefore uninformative. As above, these are proxy results on classification datasets; no deployment or safety claim is made. The corresponding code and recorded metrics are in `src/agent_selection.py` and `data/agent_selection_metrics.json`.

7 Limitations and Future Work

The benchmark is not clinical, the data are non-temporal, and only a logistic ensemble is implemented. The selective action-selection study is included, but it rests on small, largely separable classification proxies and a single seed. Future work will add interval type-2 membership functions, public physiological time series, temporal splits, distribution-shift tests, repeated seeds and confidence intervals, MC dropout and Bayesian baselines, additional selective-prediction baselines, and evaluation on datasets with genuine action ambiguity.

8 Conclusion

The proposed fuzzy-calibrated conformal baseline provides an executable and auditable starting point for adaptive uncertainty-aware AI, now including a selective action-selection protocol with abstention. Its current result is a methodological proxy, not a high-stakes deployment claim. The next stage is validation on time-dependent physiological and engineering datasets with stronger baselines and repeated trials.

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Table 2: Selective action selection on public proxies (seed 20260915, $K = 7$, $\alpha = 0.10$, $\beta = 0.50$). Lower selective risk and AURC are better.

Proxy	$\lvert\mathcal{A}\rvert$	Cov. base	Cov. fuzzy	Risk base	Risk fuzzy	AURC
digits	10	0.867	0.897	0.006	0.018	0.003
wine	3	0.944	1.000	0.000	0.000	0.000
iris	3	0.867	0.967	0.071	0.038	0.014